

Computer Vision-Based Structural Health Monitoring: Evaluation of a Lightweight YOLOv8 Model for Automated Surface Crack Detection

Rifa Ryaeesa

*Dept. of Computer Science and Engineering
East Delta University, Chattogram, Bangladesh
rayeesarifastudent9@gmail.com*

Tawsif Imroj

*Dept. of Computer Science and Engineering
East Delta University, Chattogram, Bangladesh
tawsifimroj@gmail.com*

Antu Chowdhury

*Dept. of Computer Science and Engineering
East Delta University, Chattogram, Bangladesh
antu.c@eastdelta.edu.bd*

Abstract—Structural Health Monitoring (SHM) is crucial aspect for sustaining infrastructure longevity in civil engineering and infrastructure industry. Mostly the structural monitoring depends on manual inspections. Although it may result in positive identification of crack's existence on the surface level but it may also come at the risk of delayed detection resulting in putting a mass in hazardous situation. While manual inspection may not fully be sufficient to detect the anomaly an automated detection system has the potential to identify surface level cracks without human intervention. This study proposes an automated interdisciplinary solution leveraging the deep learning lightweight YOLOv8s (Small) deep learning architecture. By training on a specialized dataset of 2,390 concrete crack images, the system achieved a Mean Average Precision (mAP@0.5) of 98.6% with an inference speed of 16.2 ms per image. These results demonstrate that high-fidelity structural assessment can be achieved on standard computational hardware, not necessarily using high computational power (Super-computer , having larger GPU) providing a scalable tool for real-time civil engineering inspections proving to be efficient for civil engineers.

Index Terms—Structural Health Monitoring, Computer Vision, YOLOv8, Deep Learning, Crack Detection, Lightweight Models.

I. INTRODUCTION

Civil infrastructure deteriorates due to environmental stressors and mechanical loading. Traditional inspection involves manual visual assessment, which is inherently slow, expensive, and limited by human fatigue. There is a significant research gap in evaluating "lightweight" deep learning models that can operate on standard hardware rather than high-end GPUs. This study implements a YOLOv8-based pipeline to bridge the gap between CSE research and practical Civil Engineering application, focusing on real-time automated crack detection.

II. LITERATURE REVIEW

Traditional SHM relied on contact-based sensors like strain gauges, which are difficult to maintain [1]. Recent advancements in Convolutional Neural Networks (CNNs) have revolutionized defect detection by learning feature hierarchies [2].

While models like ResNet and VGG provide high accuracy, they often lack the speed required for real-time field deployment [3]. The YOLO (You Only Look Once) architecture addresses this by treating detection as a single regression problem [4].

III. METHODOLOGY

A. Methodological Workflow

The research follows a sequential pipeline illustrated in Fig. 2.

B. Data Distribution Analysis

To ensure the model was trained on a spatially diverse dataset, a label analysis was performed. Fig. 1 shows the distribution of crack instances across the image domain, confirming that the dataset includes varied crack locations and sizes.

C. Implementation Details

A dataset of 2,390 images was sourced from Roboflow Universe (Crack-7rsjb). The YOLOv8s variant (11.1M parameters) was chosen for its lightweight profile. Training was conducted via Transfer Learning on a Tesla T4 GPU for 25 epochs.

IV. RESULT ANALYSIS AND DISCUSSION

A. Quantitative Performance

The model achieved high reliability across all standard metrics as summarized in Table I. The mAP@0.5 score of 98.6% exceeds typical literature baselines for lightweight architectures.

B. Qualitative Model Validation

A qualitative comparison was conducted by evaluating the model's predictions against the ground-truth annotations from the validation set. As illustrated in Fig. IV-B, the AI-generated bounding boxes (right) show near-perfect alignment with the original manual labels (left).

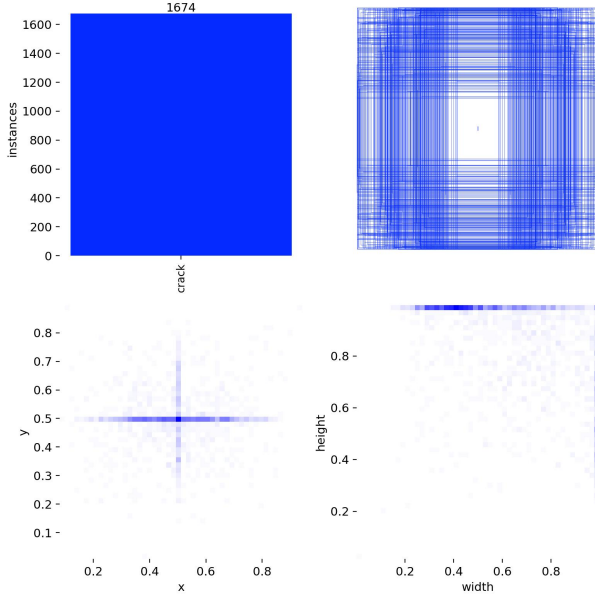


Fig. 1. Spatial distribution and box dimensions of the concrete crack dataset.

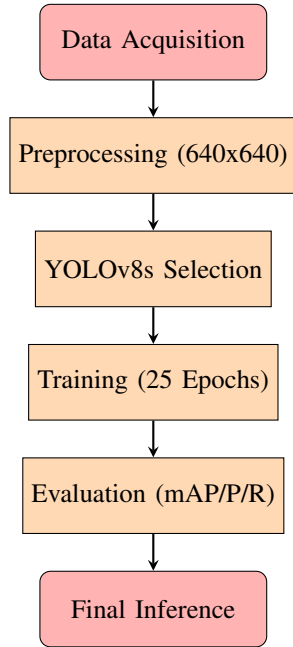


Fig. 2. Flowchart of the Proposed Automated Detection Methodology.

TABLE I
MODEL PERFORMANCE EVALUATION SUMMARY

Metric	Value	Significance
Precision (P)	95.4%	Accurate localization
Recall (R)	93.5%	High sensitivity
mAP@0.5	98.6%	Overall reliability
Inference Speed	16.2 ms	Real-time (60 FPS)

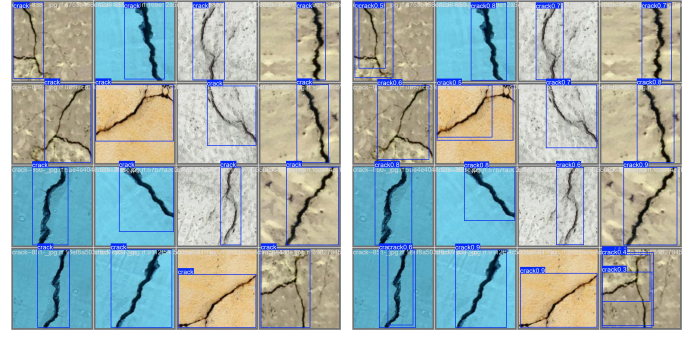


Fig. 3. Manual Labels

Fig. 4. AI Predictions

C. Scientific Visualizations

As shown in Fig. 5, the model reached convergence rapidly. The scatter distribution of labels (Fig. ??) confirms a lack of spatial bias in the training data.

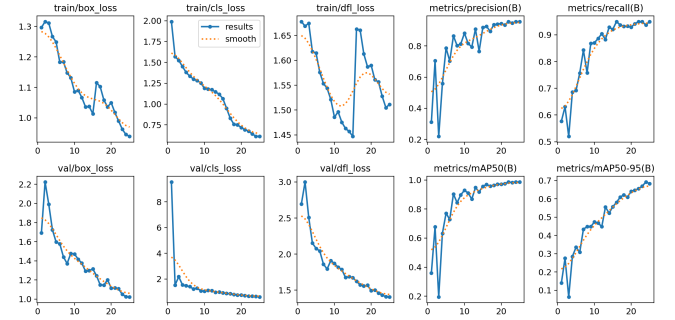


Fig. 5. Training and Validation Metrics (mAP, Precision, and Loss).

D. Qualitative Validation

Figure 6 provides visual proof of the system's performance on unseen test data. The model accurately identified multiple structural cracks with high confidence.

V. CONCLUSION

This study proves that lightweight YOLOv8 models can serve as robust, real-time tools for Structural Health Monitoring. Achieving a 98.6% accuracy with 16.2ms inference time satisfies both Civil Engineering safety requirements and CSE computational constraints. Future research will explore 3D crack depth estimation using LiDAR-integrated vision systems.

REFERENCES

- [1] C. Tello-Gil, S. Jabari, L. M. Waugh, M. Masry, and J. McGinn, "Crack detection and dimensional assessment using smartphone sensors and deep learning," *Canadian Journal of Civil Engineering*, 2024.
- [2] X. Pan, T. T. Yang, C. Ventura *et al.*, "A review of recent advances in data-driven computer vision methods for structural damage evaluation: algorithms, applications, challenges, and future opportunities," *Archives of Computational Methods in Engineering*, 2025.
- [3] P. D. Ogunjinmi, S.-S. Park, B. Kim *et al.*, "Rapid post-earthquake structural damage assessment using convolutional neural networks and transfer learning," *Sensors*, vol. 22, no. 9, p. 3471, 2022.
- [4] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics yolov8," 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>

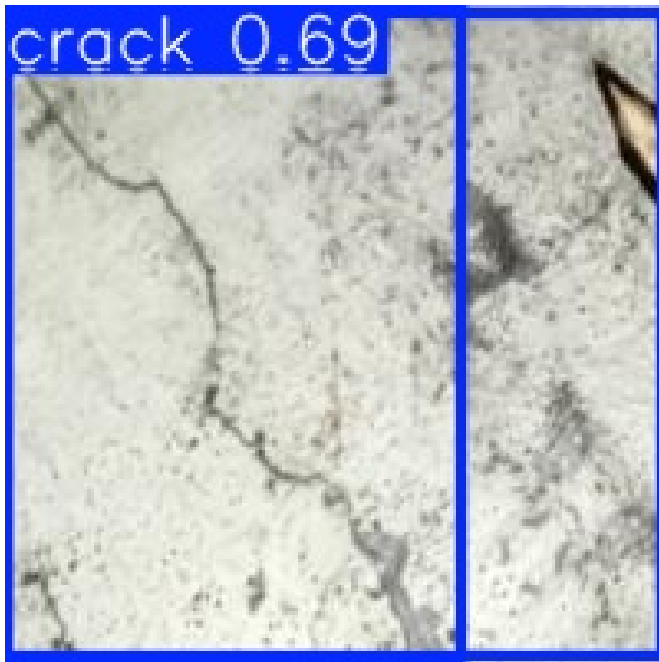


Fig. 6. Inference Result: Automated detection of multiple concrete cracks.